Data Analytics (DA)

Module -2

Power BI Assignment 1 – Data Transformation & Data Modeling

Assignment Task Overview

Problem Statement:

As a **Data Analyst**, you will work with an **E-commerce sales dataset** to perform data preparation and modeling using **Power BI**. The dataset contains multiple related files representing order-level, product-level, and sales target information.

The focus of this assignment is on **data import, transformation, cleansing, aggregation, and relationship modelling** using **Power BI and Power Query Editor**, while ensuring consistency, accuracy, and business relevance of the prepared data model.

Dataset Description :

You will work with the following three datasets:

- ***List of orders***

  Attributes: *Order ID, Order Date, Customer Name, City, State, Category, Amount, Profit*
- ***Order Details***

  Attributes: *Order ID, Category, Sub-Category, Quantity, Amount, Profit*

- ***Sales Target***

  Attributes: *Category, Month of Order Date, Target*

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Tasks

Data Import & Transformation

1. **Data Import:**
   - Import *List of Orders.csv* into Power BI.
   - Open *the list of orders* in **the Power Query Editor** using the **Transform Data** option.
   - Import *Order Details.csv* and *Sales Target.csv* into Power Query Editor.

2. **Row Limiting & Data Types:**
    - Restrict the *List of Orders* table to the **first 500 rows**.
    - Set the **Order Date** column to **the Date** data type.
    - Change the **Amount** and **Target** columns to **fixed decimal numbers.**
3. **Text Formatting:**
    - Format the **CustomerName** column to **proper case** to ensure consistent capitalization.
4. **Column Creation:**
    - Merge the **City** and **State** columns to create a new column named **Location** in the format: ***City, State***
    - Create a custom column named **Profit Margin** calculated as: ***Profit / Amount*** (expressed as a percentage)
5. **Conditional Column:**
    - Add a conditional column named **Profit Status** based on:
        - Profit < 0 → *Loss*
        - Profit = 0 → *Break-Even*
        - Profit > 0 → *Profit*
6. **Merging Data (Joins)**
    - Merge the *List of Orders* and *Order Details* tables using **Order ID**.
    - Name the merged output table as **Orders Data**.
7. **Handling Missing Data & Duplicate Data**
    - Identify missing values across all tables.
    - Define and apply appropriate strategies such as:
        - Replacing missing values
        - Removing records
        - Retaining nulls with justification
    - Identify duplicate rows and define a suitable handling strategy based on business logic.

8. **Sorting and Filtering Data**

- Sort records in the **Orders Data** table by **Order Date** in **descending order**.
- Apply filters to analyze orders from a specific state (e.g., **Tamil Nadu**).

9. **Grouping and Aggregating Data**

- Duplicate the *Order Details* table and perform the following:
    - Count of Order ID
    - Average Profit by Category
    - Total Amount by Sub-Category
- Duplicate the *Sales Target* table and aggregate the total Target by **Month of order date.**

10. Data Modeling:

- Establish a relationship between:
    - *List of Orders* and *Order Details* using **Order ID**
- Establish a relationship between:
    - *Order Details* and *Sales Target* using **Category**

2. Use **Manage Relationships** to ensure relationships are active and correctly defined.

Deliverables

- Submit a project folder containing:
    1. **Power BI File (.pbix)** that includes the following:
        - Imported datasets
        - All Power Query transformations
        - Merged tables
        - Data model relationships
    2. **PDF Document** that includes:
        - Screenshots of the solution and the data model view
        - Brief explanation of transformation and modeling logic

Skills to be Evaluated and Mapping

Assignment Section	Core Skill	Sub-Skill	Proficiency
Data Import	Data Preparation	Data Loading	Beginner
Data Transformation	Data Cleaning	Data Types, Formatting	Beginner
Calculated Columns	Business Logic	Custom & Conditional Columns	Beginner
Data Merging	Data Integration	Joins in Power Query	Intermediate
Data Quality	Data Validation	Null & Duplicate Handling	Intermediate
Sorting & Filtering	Data Exploration	Filters & Sorting	Beginner
Aggregation	Data Analysis	Grouping & Aggregates	Intermediate
Data Modeling	Data Modeling	Relationship Management	Intermediate

Rubric & Mark Allocation (Total 10 Marks)

Skill / Component	Marks	Criteria for Full Marks	Partial Credit Description	Partial Marks Range
Data Import & Transformation	3	Correct transformations, data types, and columns	Minor transformation errors	1-2
Aggregation & Filtering	3	Correct grouping, sorting, and filtering	Partial logical issues	1-2
Data Merging & Quality	2	Accurate joins and data handling	Join or data issues	0.5-1
Data Modeling	2	Correct and active relationships	Incorrect/inactive relationships	0.5-1

DA Skill Taxonomy Structure

- **Data Preparation:** Data import, Power Query transformations
- **Data Cleaning:** Formatting, null handling, deduplication
- **Data Aggregation:** Grouping and summarization
- **Data Integration:** Table merging and relationships
- **Data Modeling:** Schema design and relationship management
- **Business Interpretation:** Translating business logic into data models

Evaluation Guidelines

- Verify all datasets are correctly imported.
- Validate Power Query steps and transformation logic.
- Check calculated and conditional columns for correctness.
- Review relationship design and model integrity.
- Assign full or partial marks based on accuracy and clarity.
- Provide feedback highlighting strengths and improvement areas.

Achievement Criteria & Passing Rules

- At least **partial (non-zero) marks** must be obtained in every section.
- Students must demonstrate proficiency in **Power BI data transformation and modeling**.
- Business logic must be accurately translated into Power Query steps.
- Clear documentation and screenshots are mandatory for evaluation.